

Grow | Innovate | Sustain

VIEM PROJECT

Circular Smart Farming System and First AI Tetum Assistance

Vorakai Konekta Challenge 2026

www.viemproject.com

VIEM



Grow Smarter
Go Further

About VIEM

VIEM is a modern agribusiness initiative committed to building a smarter, more regenerative, and technology-enabled agricultural ecosystem in Timor-Leste.

We combine circular agriculture, controlled farming systems, IoT technology, automation, and data to improve agricultural productivity while making better use of limited resources.

Our goal is to create a farming model that is productive, measurable, resource-efficient, and replicable for Timor-Leste.

Our Vision

To become a leading modern and regenerative agribusiness company, creating a productive, circular, and sustainable agricultural ecosystem for Timor-Leste.

Our Focus

- Integrating technology and data to improve farming efficiency and productivity.
- Transforming organic waste and agricultural by-products into valuable resources, including animal feed.

The Problem
in Timor Leste

1 Agriculture Should Waste Less and Produce More

The Production depends on several resources that are often treated separately:

- Water
- Feed
- Nutrients
- Waste
- Labor
- Information

In **conventional systems**, resources frequently move in a **linear direction:**

INPUT → PRODUCTION → OUTPUT → WASTE

VIEM sees an opportunity to redesign this relationship. Instead of continuously adding inputs and discarding outputs, **agricultural activities can be interconnected** so that **the output** of one process **becomes an input** for another.

2 Farming Should Be Smarter and Connected

At the same time, farmers and operators often depend on manual observation to understand farm conditions. Without continuous monitoring:

- Farm conditions can change unnoticed
- Equipment failures may be detected too late
- Production records can become inconsistent
- Resource consumption is difficult to measure

Agriculture in Timor Leste

According to the Food and Agriculture Organization of the United Nations (FAO)

(https://www.fao.org/timor-leste/our-office/timor-leste-at-glance)

70%
of the Population

Agriculture, livestock production, forestry, fisheries and aquaculture provide livelihoods for around 70% of Timor-Leste's population.

80%
of Rural Households

Around 80% of rural households engage in crop and livestock production or artisanal fisheries.

16.7%
GDP Contribution

Agriculture accounted for approximately 16.7% of Timor-Leste's GDP in 2023.

Agriculture provides **livelihoods for a large share of Timor-Leste's population**, yet its **contribution** to national economic output remains **relatively limited**

This highlights the **opportunity to improve** productivity, resource efficiency, and agricultural value creation

Our Question

How can we **PRODUCE MORE** with **LESS LAND, LESS WASTE,** and **SMARTER USE OF RESOURCES?**

Our Answer

VIEM Circular Smart Farming System

An **integrated farming ecosystem** designed to produce more from limited space, transform waste into valuable resources, and use technology to manage **farming more efficiently**.
We combines circular agriculture, modern cultivation systems, IoT, automation, and data into **one connected agricultural ecosystem.**

Produce More

Intensive & Controlled Farming using **Aquaponics and hydroponics** enable **more productive cultivation** within limited farming space while providing **greater control** over growing conditions.

Waste Less

Circular Resource System started with **Organic waste** that converted through BSF/maggot cultivation into **valuable resources**, including feed, while agricultural by-products are reintegrated wherever technically appropriate.

Use Resources Smarter

IoT, Automation & Data that using Sensors continuously **monitor farming conditions**, while local automation and a web-based system help manage water, nutrients, equipment, and production data **more efficiently**.

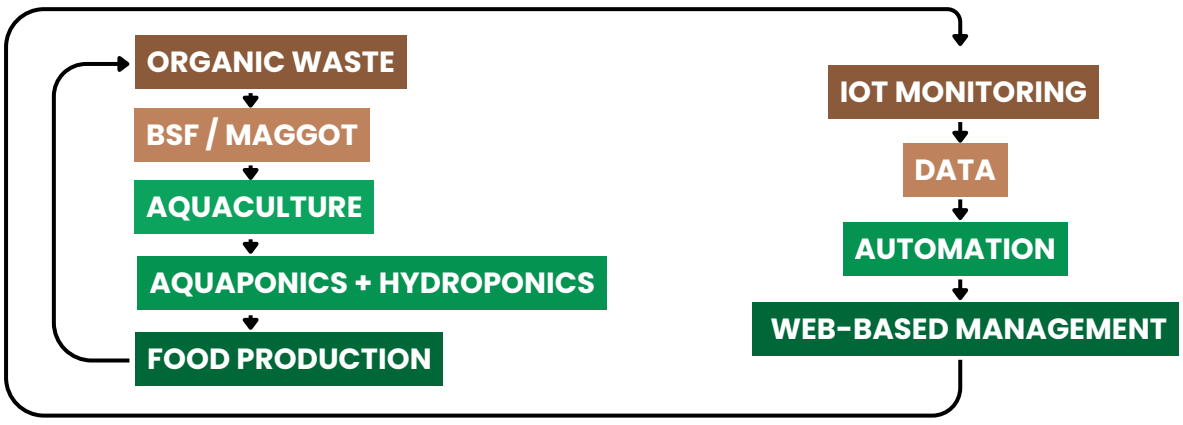


Fig. 1 *VIEM Circular Smart Farming Ecosystem*

Produce More | Waste Less | Farm Smarter

How The System Works

From Circular Farming to Smart Farming

VIEM connects physical farming operations with digital technology to create a system that can monitor, respond, record, and continuously improve.

1 Sense for Real-Time Monitoring

Sensors continuously collect critical farm data such as:
pH, EC/TDS, Water Temperature, Air Temperature, Humidity, and Water Level, giving operators **better visibility** into actual farm conditions.

2 Local Automation

The system can **automatically control** essential equipment based on predefined conditions such as Water Pump, Aerator, Irrigation, Water Circulation and liquid fertilizer injection.
With this system, critical functions **can continue locally** without depending entirely on internet connectivity.

3 Starlink Satellite Connectivity

VIEM uses **Starlink satellite connectivity** to connect farm operations with the VIEM digital platform, **enabling data synchronization and remote** access even in locations where conventional internet infrastructure may be limited.

4 Web-Based Management

Farm data is transformed into information that **operators can understand** and use for Live Monitoring, Alerts, Historical Data, Production Records, Resource Tracking and Farm Control.

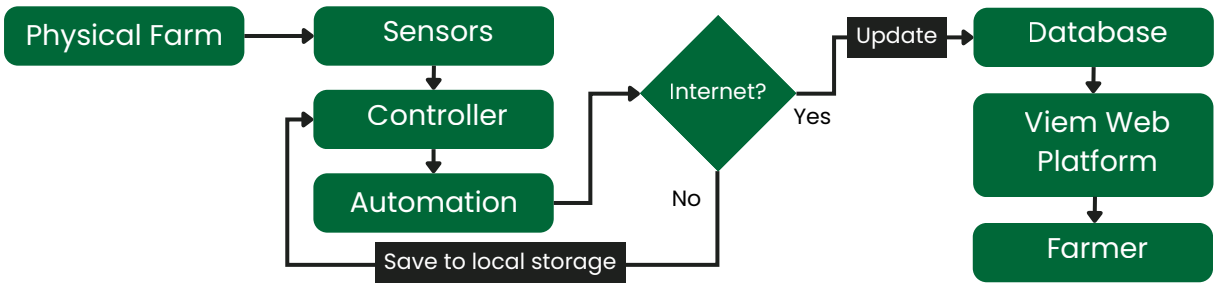
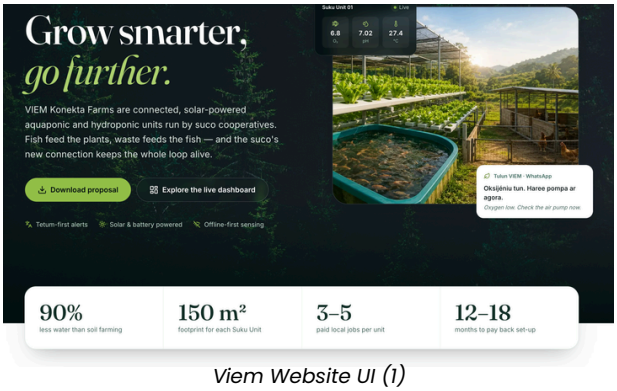


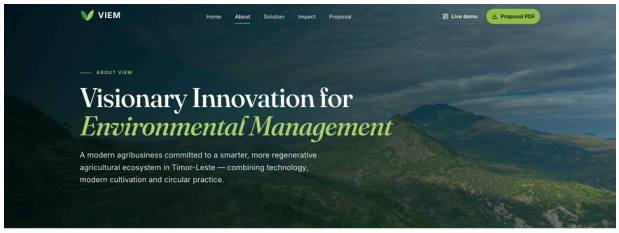
Fig. 2 *VIEM Smart Farming System Architecture*

The farm keeps **operating locally** **Connectivity** makes it visible, measurable, and **smarter**

Overview

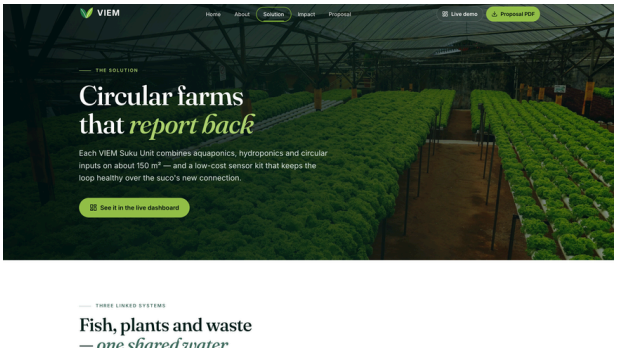


Viem Website UI (1)

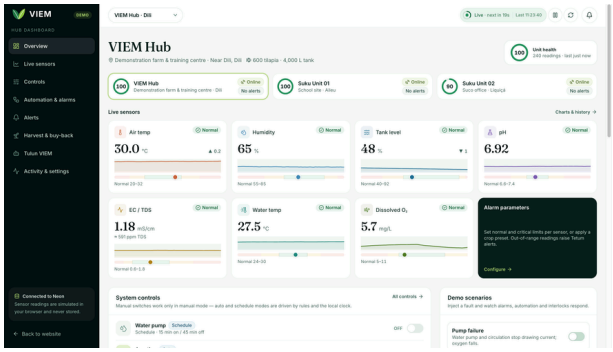




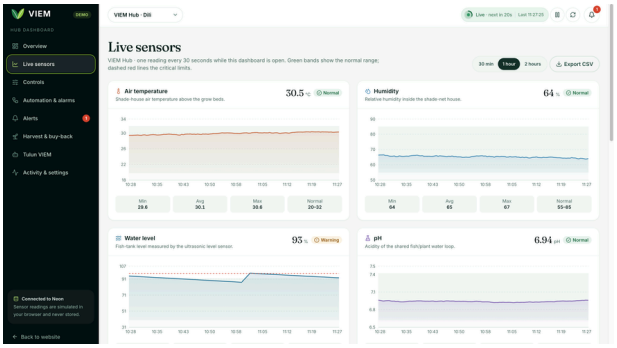
Viem Website UI (2)



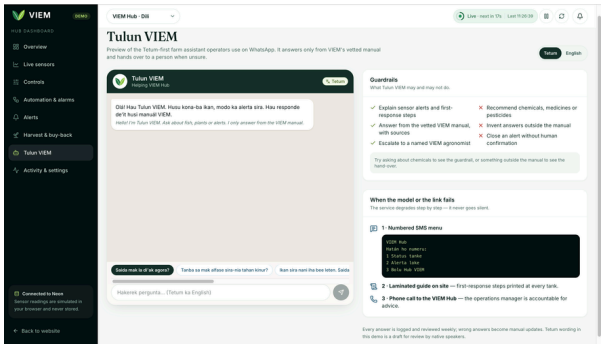
VIEM Website UI (3)



Dashboard General Overview



VIEM Live Sensor



VIEM First Tetum Farm Assistant

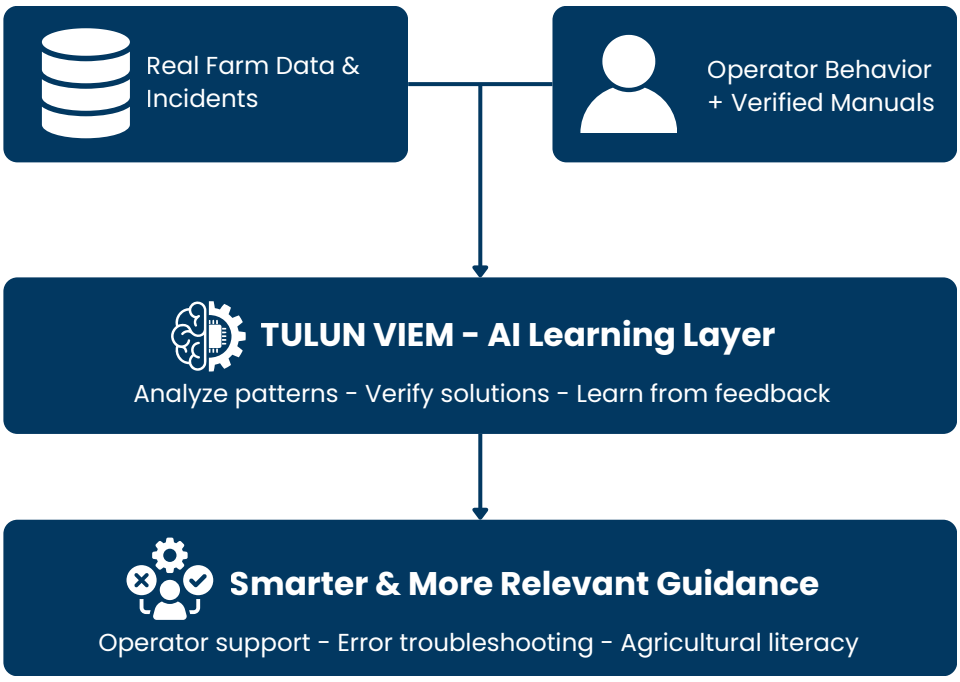
Fig. 3 *VIEM Web Platform Prototype*

[Click here for Website Demo]

“TULUN VIEM “
AI Agricultural Assistant

Tulun VIEM uses **Gemini AI** and **real farm data** to provide practical agricultural guidance. By analyzing sensor readings, incident histories, operator behavior, and verified manuals, it helps farmers troubleshoot problems and make informed decisions.

Through human-verified continuous learning, resolved incidents and expert-approved solutions enrich its knowledge base, enabling smarter, more accurate, and locally relevant guidance over time.



Layer	Technology	Primary Function
Interface	VIEM Web, Multilingual UI	Farmer interaction & guidance
Intelligence	Gemini API, RAG	Contextual AI responses
Knowledge	VIEM Database, Verified SOPs	Trusted data & learning
Operations	ESP32, IoT, Local Controller	Monitoring, Farm Control Trigger & Automation

Fig. 4 *TULUN VIEM Architecture*

Human-Guided, Not AI-Controlled

Tulun VIEM is designed as a Tetum-first learning and troubleshooting companion. Its initial role is to explain sensor readings, clarify alerts, guide operators through documented checks, and improve agricultural literacy. AI provides advice; authorized operators remain responsible for decisions

Stage	Operator-facing behavior	Evidence captured
Observe	Read available farm data, incident logs, questions and alert responses	Sensor values, timestamps, issue descriptions
Compare	Retrieve relevant SOPs, agricultural references and locally reported practices	Sources cited, suggested checks and alternatives.
Verify	An operator checks the cause, action taken and actual outcome	Resolution status, reviewer and notes
Learn	Add approved cases to a searchable knowledge base for similar incidents	Verified case history; not automatic model retraining

Verified cases are added to the knowledge base, allowing Tulun VIEM to provide more relevant guidance when similar situations occur.

AI Safety & Human Oversight Framework

Principle	Implementation
Human Decision Authority	Operators approve actions; AI cannot independently control equipment or close alerts.
Verified Knowledge	Agricultural guidance is grounded in approved manuals and expert-reviewed cases.
Safety Boundaries	No chemical or pesticide recommendations; uncertain or high-risk cases require human review.
Data Protection	Secure server-side API access, validated records, and controlled data permissions.
Offline Resilience	Critical farm automation continues locally, independent of AI or internet connectivity

VIEM Pilot Project

VIEM will begin with a small-scale integrated pilot **to test** how circular farming, controlled cultivation, and smart technology perform together under real operating conditions. This is an overview for our farm integration and validation model:

Farm Integration

1 Production System

Integrated farming systems designed to produce fish and vegetables efficiently within limited space.

- **Aquaculture** — Fish Production
- **Aquaponics** — Fish natural fertilizer for plant production
- **Hydroponics** — Controlled plant production

2 Circular System

Turn suitable organic waste into productive inputs and reduce resource waste.

- **Organic Waste** — Food waste, Agricultural by products
- **BSF/ Maggot** — Waste conversion into valuable resources
- **Feed** — Animal and Fish feed

3 Smart System

Turn suitable organic waste into productive inputs and reduce resource waste.

- **Sensors**
- **Local Controller**
- **Auto-mation**
- **Starlink**
- **VIEM Platform**

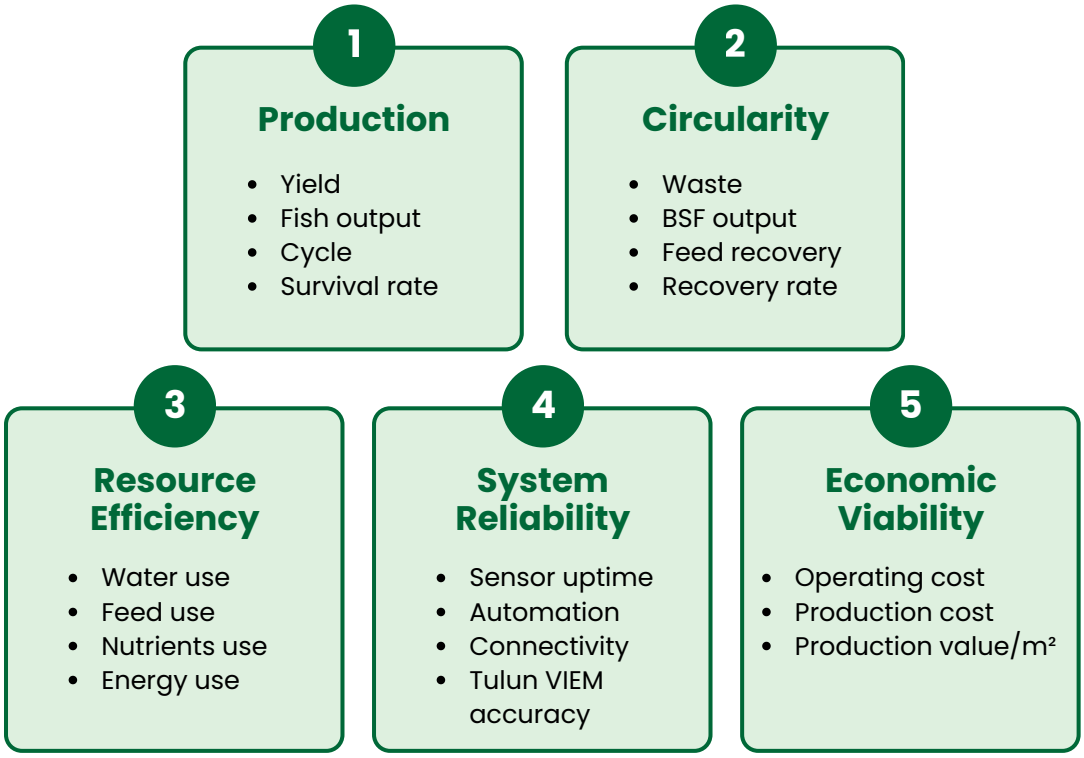
Validation Model

1. **Build** — Build the small-scale farming and IoT prototype
2. **Integrate** — Connect production, circular resources, sensors, and automation
3. **Test** — Operate the system under real farming conditions
4. **Measure** — Collect production, resource, operational, and system data
5. **Improve** — Use pilot results to refine the system before replication
6. **Deploy** — Deploy the validated model for real-world implementation.

Fig. 5 *Integration and Validation model*

Measuring What Matters

VIEM **evaluates** the pilot across five key dimensions to determine whether the system is productive, resource-efficient, reliable, and ready for further deployment.



Dimension	Measurement Method
Production	Harvest records, weight measurements, production logs
Circularity	Input and output weight records
Resource Efficiency	Meter readings, consumption records
System Reliability	System logs, outage tests, incident records, expert review, operator feedback
Economic Viability	Expense tracking, cost and revenue analysis

We measure success not only by whether the technology works, but by whether the entire **farming system creates measurable agricultural value.**

Designed for Deployment, Not Just Demonstration

VIEM is designed for real-world deployment—where connectivity can fail, power can be interrupted, equipment can malfunction, and operating conditions can change. Here is how we plan to address these challenges:

Technical Failure

What Could Fail?	VIEM Response
Starlink / Connectivity Loss	Local automation continues and data is stored locally for later synchronization
Sensor Failure / Incorrect Reading	Flag abnormal readings and allow manual measurement and verification
Pump / Aerator Failure	Trigger alerts and allow manual operation or equipment replacement
Power Outage	Maintain safe operating procedures and prioritize critical equipment; backup power can be added based on site requirements.

Financial Failure

What Could Fail?	VIEM Response
High Initial Investment	Start with a small modular pilot and expand only after validation
Operating Cost Too High	Measure energy, feed, water, and connectivity costs during the pilot
Low Production Value	Compare production value against operating costs before deployment

Operational & Technical Failure

What Could Fail?	VIEM Response
Limited Technical Skills	Simple interfaces, SOPs, and operator training, Provide Tetum first AI Agriculture Advisor
Incorrect Operator Input	Validation, confirmation steps, and manual override
Low Technology Adoption	Keep essential farm operations possible locally and involve operators during pilot testing
Language Barrier	Develop farmer-facing support with Tetum-first communication

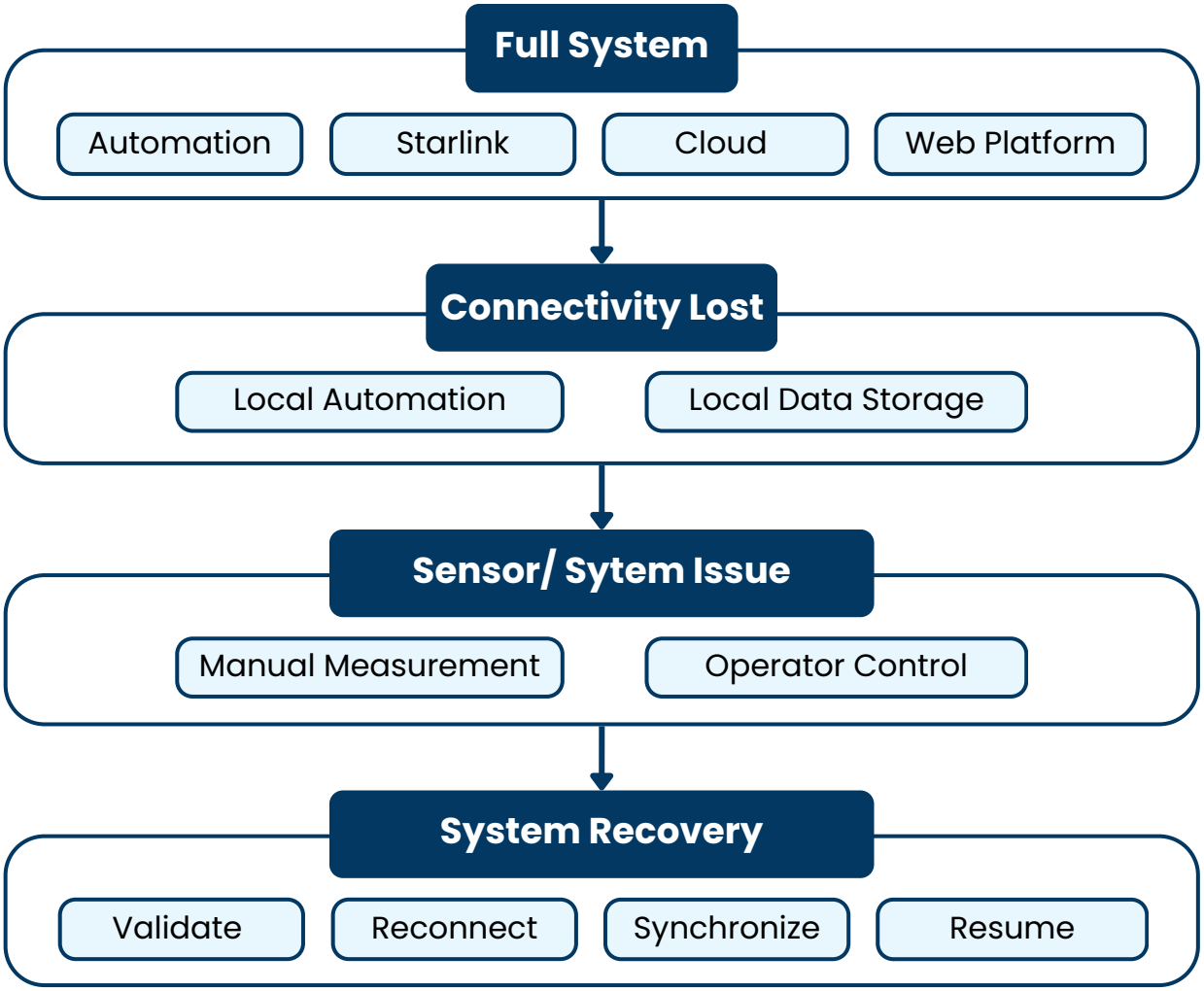


Fig. 6 *Failure & Recovery Model*

1 Month Build Sprint

From Concept to Working Prototype

Week 1	Week 2	Week 3	Week 4
Hardware	Platform	Resilience	AI & Demo

W1

Hardware & Sensor Setup

Connect ESP32 controllers with selected water and environmental sensors. Establish reliable data collection and basic sensor validation

W2

VIEM Platform Integration

Connect sensor data to the existing VIEM web platform. Develop basic monitoring views, alert notifications, and historical records

W3

Local Automation & Connectivity Testing

Implement basic local control for selected equipment and test operation during simulated internet outages. Validate local data storage and synchronization after reconnection

W4

Tulun VIEM & Final Demonstration

Integrate an initial Gemini powered assistant with selected farm data and verified agricultural references. Test basic Tetum guidance and demonstrate the connected prototype

Expected Outcome

IoT Monitoring
Sensor readings displayed on VIEM Web

Offline Resilience
Local control during connectivity outages

Tulun VIEM
Initial AI guidance using verified references

Pilot Readiness
Test findings and next development priorities

Our **30 day goal** is to demonstrate a working, connected farming prototype, not to complete an entire farm. The Build Sprint will validate the core technology and identify improvements before deploying VIEM's integrated circular farming pilot in Laleia, Timor Leste.

Deployment Roadmap

From Pilot to Real Operations

A 6-month pathway to build, validate, and transition VIEM from pilot testing into real agricultural operations

Phase	Duration	Plan	Outcome
1	Month 1	**Deploy Pilot Project** Build the first integrated system combining farming, circular resources, IoT, and automation.	Operational Pilot & Baseline Data
2	Month 2–3	**Test & Learn** Measure production, resource efficiency, reliability, and economic viability	Validated Performance Data
3	Month 4	**Improve the System** Use field data to improve cultivation methods, resource flows, automation, and operating costs	Optimized Farm System
4	Month 5	**Validate in Real Operations** Run the improved configuration and verify technical, operational, and production performance.	Validated Pilot Model
5	Month 6	**Transition to Real Production** Move from pilot validation into continuous VIEM agricultural operations based on validated results	Operational VIEM Farm

Each phase progresses **based on evidence** from the previous stage, **not assumptions**

We Are Not Starting From Zero

What Viem Already has

1

± 700m² Land Bank

Laleia - Timor Leste

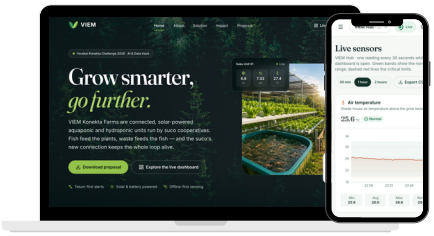


- ± 700m² available land for pilot development
- Suitable for integrated farming system
- Land access already available for VIEM

2

VIEM Web Platform

Digital Infrastructure



- Web based farm control
- Production records & resource tracking
- Existing prototype ready for integration

3

System Design

Integrated Concept



- Circular farming architecture
- IoT system design
- Pilot validation & deployment framework

Land + **Platform** + **System Design** → **Ready for Development**

VIEM is not starting from an idea alone.
The land, digital platform, and system design are already in place.
The next step is to integrate them into a working pilot.

What We Need to Make It Happen

Category	What we Need	Purpose
Farm Infrastructure	• Greenhouse, • hydroponics, • aquaponics, • aquaculture tanks, • BSF unit, water • circulation & filtration	Build the physical production and circular farming system
IoT & Automation	• Sensors • ESP32/local controllers • pumps, aerator • actuators, • electrical & control components	Connect farm operations with the VIEM platform and automate essential functions
Connectivity & Resilience	• Starlink • local network • local data storage • electrical installation & essential backup power	Maintain reliable monitoring and local system operation
Farm Inputs	• Seeds • fish stock • feed, nutrients • growing media • initial production inputs	Start the first production cycles
Pilot Operations	• Labor • maintenance • testing • monitoring • initial operational expenses	Operate and validate the pilot under real conditions
Contingency	• Logistics • price variation • unexpected implementation costs	Cover unexpected costs and keep pilot implementation on schedule

What Each Budget Unlocks

Category	Lean Pilot	Vorakai Pilot	Full Pilot
Farm Infrastructure	$3,500	$4,500	$6,500
IoT & Automation	$1,200	$1,800	$2,500
Connectivity & Resilience	$800	$1,200	$1,800
Farm Inputs	$600	$800	$1,200
Pilot Operations	$900	$1,200	$2,000
Contingency & Logistics	$500	$500	$1,000
TOTAL	**$7,500**	**$10,000**	**$15,000**

LEAN PILOT

Minimum Viable Validation

- Smaller production capacity
- Essential IoT monitoring
- Basic automation
- Core farming systems
- Focus on technical validation

Prove that the integrated system works.

VORAKAI PILOT

Recommended Configuration

- ±200 m^2 pilot footprint
- Integrated farming systems
- IoT monitoring & automation
- Starlink connectivity
- Initial operational validation

Build, operate, and validate the first integrated VIEM pilot

FULL PILOT

Extended Validation

- Higher production capacity
- More complete automation
- Stronger system resilience
- Longer operational runway
- Broader performance validation

Validate the system at a stronger operational scale

WHAT SUCCESS LOOKS LIKE

For us, Success means turning technology and connectivity into real agricultural value creating a viable operation, strengthening local capability, and building evidence for sustainable growth in Timor Leste.

01 BUSINESS IMPACT
A Viable Agricultural Operation

- Validate production cost and operating economic
- Generate a high value from agricultural production
- Measure operational efficiency with technology
- Create evidence for further VIEM investment

From funded pilot to economically informed operation.

02 LOCAL IMPACT
Skills & Local Opportunity

- Build practical smart-farming skills for local operators
- Create opportunities around farm operations and maintenance
- Increase local agricultural production capacity
- Make digital farming tools accessible through tetum-first support

Connectivity creates practical value for people on the ground.

03 VORAKAI IMPACT
Connectivity creates value

- Demonstrate the value of Starlink beyond basic internet access
- Create a practical agricultural use case for connected rural communities
- Generate field evidence on IoT, monitoring, and satellite connectivity
- Show how connectivity can support productive economic activity

From connecting a site to enabling what the site can produce.

04 SCALE POTENTIAL
For what comes next

- Document what worked and what did not
- Identify digital components that can support other connected agricultural sites
- Adapt the system to different budgets, power conditions, and farming needs
- Generate evidence for future partnerships and investment

A validated use case that can be adapted and scaled.



WHERE **CONNECTIVITY**
TURNS INTO **PRODUCTIVITY**

VIEM PROJECT

VIEM Core Team



Deviana Sarmento

VIEM Director

Finance and operations professional with experience in Timor-Leste's service sector, leading VIEM's financial planning, local operations, and business sustainability

Yuma Rafi

Commissioner

Business and operations professional experienced in building efficient systems across Indonesia and Timor-Leste, supporting VIEM's strategy, business model, and operational development





Escolastico Sarmento

General Manager

Marketing professional with experience across Indonesia and Timor-Leste, leading VIEM's market development, partnerships, and local commercial opportunities

Luis Ferdinand

IT Manager

IT professional specializing in IoT integration and web-based applications, leading VIEM's farm monitoring, automation, and data-driven technology systems



*We **grow** with **purpose**,*
*we **innovate** with **vision**,*
*we **move forward** with **impact***

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